

TAPPAHANNOCK-ESSEX COUNTY AP, VA, USA

WMO: 720501

Lat: 37.850N

Lon: 76.883W

Elev: 41

StdP: 100.83

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00155

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-7.0	-22.3	0.5	-4.1	-18.3	0.7	-1.5	8.5	6.4	7.6	6.3	1.4	300	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	35.1	24.5	33.3	23.7	32.3	23.4	26.7	31.0	25.9	30.0	25.3	29.2	2.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.3	28.1	25.0	20.2	27.5	24.1	19.1	26.7	83.8	31.1	80.5	29.9	77.3	29.1	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.9	5.2	-15.1	37.2	4.2	1.7	-18.2	38.4	-20.7	39.4	-23.0	40.4	-26.1	41.7		
			-15.6	27.6	4.2	1.8	-18.6	28.8	-21.0	29.9	-23.4	30.9	-26.4	32.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	2.6	4.9	8.5	14.6	19.7	24.0	26.1	24.8	21.9	15.6	9.2	5.8
	DBStd	9.30	5.69	5.96	5.63	5.00	4.35	3.20	2.73	2.52	3.65	4.65	4.79	5.20
	HDD10.0	749	238	166	99	13	0	0	0	0	0	8	71	154
	HDD18.3	2137	488	378	308	136	39	2	0	0	9	111	277	389
	CDD10.0	2517	8	22	53	149	302	419	498	458	356	181	46	24
	CDD18.3	864	0	1	3	23	82	172	240	201	116	26	2	1
	CDH23.3	7968	0	5	31	239	704	1625	2458	1777	926	196	6	1
(14) Wind	WSAvg	1.9	2.2	2.4	2.6	2.6	2.0	1.7	1.5	1.3	1.5	1.8	1.9	1.9
	PrecAvg	1118	82	71	106	79	98	92	119	111	111	91	80	87
	PrecMax	1582	165	167	221	143	231	159	227	341	323	174	236	212
	PrecMin	830	31	20	4	34	26	33	34	35	20	0	6	26
	PrecStd	177	32	40	46	30	52	35	57	68	78	53	52	39
	DB	20.6	23.9	26.4	31.4	32.9	37.1	37.8	35.2	34.5	30.5	23.6	22.5	
	MCWB	16.0	16.5	17.0	19.6	21.9	24.9	24.8	25.3	22.8	21.8	17.7	18.4	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	17.4	20.6	23.7	28.6	31.2	33.8	36.0	33.1	32.5	27.8	21.6	20.1	
	2% MCWB	13.3	14.4	15.3	17.9	21.3	23.4	24.4	24.7	23.0	21.0	16.3	17.5	
	5% DB	14.9	17.6	21.1	26.2	29.4	32.3	33.9	32.1	30.3	25.3	19.8	17.5	
	5% MCWB	11.4	12.4	14.1	17.2	21.6	22.9	24.1	24.3	23.1	19.8	15.5	15.3	
	10% DB	11.5	14.8	18.0	23.7	27.5	30.7	32.4	30.7	28.2	23.0	17.5	14.9	
	10% MCWB	7.7	10.4	12.7	16.5	20.5	22.5	23.8	23.7	21.8	18.1	13.9	12.4	
	WB	17.7	19.0	19.0	22.6	25.2	26.6	28.6	27.2	26.4	24.8	20.2	20.7	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	19.4	22.3	22.5	25.9	29.1	32.8	32.8	31.2	29.7	26.8	22.0	21.2	
	2% WB	14.8	16.2	17.4	20.6	23.9	25.5	27.1	26.4	25.4	23.2	18.4	17.7	
	2% MCDB	16.7	18.8	21.1	25.0	28.1	30.5	31.7	30.2	28.3	25.6	20.5	18.8	
	5% WB	11.6	13.5	15.8	19.0	22.8	24.6	26.1	25.8	24.6	21.4	16.2	15.8	
	5% MCDB	14.6	16.9	19.5	23.9	26.5	29.4	30.3	29.5	27.2	23.8	18.3	17.2	
	10% WB	8.8	10.5	13.6	17.6	21.7	23.9	25.2	25.2	23.7	19.8	14.5	12.7	
	10% MCDB	11.1	13.8	17.4	22.0	25.2	28.3	29.6	28.7	26.3	22.5	17.1	14.2	
(35) Mean Daily Temperature Range	MDBR	9.9	10.8	11.3	12.7	11.4	11.3	11.0	10.6	10.9	11.6	11.2	9.8	
	5% DB	14.1	14.4	15.1	14.5	12.9	13.0	12.8	11.9	12.6	13.1	13.3	11.8	
	5% MCWBR	11.3	9.4	8.5	6.6	5.5	4.9	4.2	4.5	4.9	6.9	8.4	9.8	
	5% WB	13.1	13.4	12.4	12.2	10.7	11.5	11.1	10.2	8.9	10.4	10.5	11.6	
	5% MCWBR	11.9	10.0	8.6	7.1	5.3	4.9	4.6	4.6	4.4	6.6	8.3	10.5	
(40) Clear-Sky Solar Irradiance	taub	0.304	0.316	0.348	0.390	0.428	0.482	0.509	0.506	0.409	0.354	0.321	0.307	
	taud	2.526	2.480	2.429	2.325	2.252	2.144	2.088	2.081	2.334	2.491	2.541	2.551	
	Ebn at Noon	886	917	914	889	858	808	782	774	839	862	858	858	
	Edh at Noon	77	91	106	125	137	153	160	158	116	90	76	71	
(44) All-Sky Solar Radiation	RadAvg	2.25	3.04	4.04	5.09	5.79	6.31	6.07	5.46	4.43	3.43	2.55	1.99	
	RadStd	0.18	0.33	0.33	0.38	0.58	0.45	0.30	0.28	0.45	0.39	0.21	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+85

Nomenclature: See separate page